Divyansh Malik

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EDUCATION

Indian Institute of Technology Jodhpur, India **CGPA: 9.15/10.0**
B.Tech in Artificial Intelligence and Data Science *2025 – 2029 (Expected)*

Ramjas School, Pusa Road, New Delhi, India **Class XII: 93.0% | Class X: 97.2%**
Central Board of Secondary Education (CBSE)

RESEARCH INTERESTS

Machine Unlearning, AI Privacy, Trustworthy AI, Membership Inference Attacks, Differential Privacy

KEY PROJECTS

Auditing Class Unlearning in ResNet-18 Beyond Output Accuracy **[GitHub Repo]**
Independent Work | PyTorch, scikit-learn

- Compared Gradient Ascent, Bad Teacher distillation, and Fisher-information pruning for forgetting one CIFAR-10 class (ResNet-18, 11.17M parameters) against a retrain-from-scratch oracle, using output (accuracy, JS divergence), latent (frozen 512-d linear probe), and relearning (300-step fine-tuning attack) audits
- Showed that $<1\%$ test forget accuracy hides retained information: Bad Teacher's probe separates the forgotten class **better** than the original model (**93.96%** vs. 92.52% balanced acc.), and Gradient Ascent/Bad Teacher recover **75%** forget accuracy in 120–150 fine-tuning steps (ReAUC 0.78–0.85), while the oracle does not
- Controlled for the BatchNorm evaluation artifact (Kalani et al.) via exact retain-set recalibration (Bad Teacher train forget acc. $8.20\% \rightarrow 2.02\%$); ablated Fisher pruning with hard masking vs. unlocked healing, where the unlocked variant matched the oracle's probe (**90.10%** vs. 90.11%) and cut ReAUC from 0.191 to **0.058**, at a 2.5-pt retain-accuracy cost

Lightweight Potato Leaf Disease Classification via Feature Distillation July 2026 – Present
Supervisor: Dr. Pratik Mazumder (IIT Jodhpur) | PyTorch, torchvision, scikit-learn

- Designed a 238K-parameter (at inference) CNN (CBAM attention, depthwise-separable convolutions) for 7-class potato disease classification on an uncontrolled-condition dataset (3,076 images)
- Distilled intermediate features from a fine-tuned EfficientNet-B0 teacher (4.02M parameters) into the student via a learned 1×1 projection head and cosine-alignment loss; the projector is training-only, so inference cost is unchanged
- Reached **79.55%** accuracy (**0.791 macro F1**) on a stratified 616-image hold-out set, retaining about 90% of the teacher's accuracy (88.8%) with a **16.9× parameter reduction**

Full-Stack Real-Time Messaging & Call Platform (Backend Architecture) **[Live Demo] | [GitHub Repo]**
DevlUp Labs, IIT Jodhpur | Node.js, Express.js, MongoDB, JWT, Arcjet, Cloudinary

- Engineered a RESTful communication API with **JWT** (`HttpOnly` cookie) authentication and `bcryptjs` password hashing, supporting media messaging via **Cloudinary**, threaded replies, and chat archiving
- Architected MongoDB schemas for voice/video call logging and granular access control, differentiating user-level soft deletes from sender-authorized hard deletes
- Hardened API endpoints against automated abuse and credential stuffing via **Arcjet** security middleware, enforcing sliding-window rate limiting (**100 req/min**) and spoofed-bot mitigation

RELEVANT COURSEWORK

- **Mathematics & Machine Learning:** Linear Algebra & Differential Equations, Multivariable Calculus & Real Analysis, Introduction to Machine Learning*, Probability, Statistics & Stochastic Processes*, Discrete Mathematics*
- **Computer Science & Systems:** Data Structures & Algorithms*, Principles of Computer Systems*, Introduction to Computer Science

**Autumn 2026 semester (completion: November 2026)*

TECHNICAL SKILLS

- **Languages:** Python, C++, C, JavaScript, LaTeX
- **Machine Learning & Data Science:** PyTorch, torchvision, scikit-learn, NumPy, Pandas, Matplotlib
- **Systems, Web & Developer Tools:** Linux/Ubuntu, Git, Node.js, Express.js, React.js, MongoDB, Postman

SCHOLASTIC ACHIEVEMENTS

- **JEE Advanced & JEE Main 2025:** Ranked in the **top 1%** among over 1.4 million candidates nationwide
- **Indian Olympiad Qualifier in Mathematics (IOQM 2023–24):** Awarded Certificate of Merit by the Mathematics Teachers' Association (India) for clearing the national qualification threshold (Stage 1 of the International Mathematical Olympiad)
- **State Science Talent Scholarship (MVPP / JSTSE):** Awarded by the Directorate of Education, Govt. of NCT of Delhi, for ranking in the state-level Junior Science Talent Search Examination